```
Product-Wise Sales With Product Name

select *
from `e1.order_items`
limit 5

select *
from `e1.products`
limit 5

select
  p.ProductID,
  p.ProductName,
  sum(oi.Total) as total_sales
from `e1.products` as p
inner join `e1.order_items` as oi
on  p.ProductID = oi.ProductID
group by p.ProductName, p.ProductID









CategoryName-Wise total Sales

select *
from `e1.categories`
limit 5
```

```sql
select
    c.CategoryID,
    c.CategoryName,
    sum(o.Total) as total_sales
from `e1.products` as p
inner join `e1.order_items` as o
on p.ProductID = o.ProductID
inner join `e1.categories` as c
on c.CategoryID = p.CategoryID
group by c.CategoryID, c.CategoryName
order by total_sales desc;
```

Calculate total sales generated by products from each supplier

```sql
select *
from `e1.suppliers`
limit 5

select
    s.SupplierID,
```

```
    s.SupplierName,
    sum(o.Total) as total_sales
from `e1.products` as p
inner join `e1.order_items` as o
on p.ProductID = o.ProductID
inner join `e1.suppliers` as s
on s.SupplierID = p.SupplierID
group by s.SupplierID, s.SupplierName
order by total_sales desc;


Display all products and any order-item sales associated with
them.

select
    p.ProductID,
    p.ProductName,
    sum(o.Total) as total_sales
from `e1.products` as p
inner join `e1.order_items` as o
on p.ProductID = o.ProductID
group by p.ProductID, p.ProductName
order by total_sales desc




Find Products With No Sales
```

```sql
select
  p.ProductID,
  p.ProductName,
  sum(oi.Total) as sales
from `e1.products` as p
left join `e1.order_items` as oi
on p.ProductID = oi.ProductID
group by p.ProductID, p.ProductName

select
  p.ProductID,
  p.ProductName,
  sum(oi.Total) as sales
from `e1.products` as p
left join `e1.order_items` as oi
on p.ProductID = oi.ProductID
where oi.ProductID is null
group by p.ProductID, p.ProductName


Customers who registered but have never placed an order.

select
  c.Name,
  c.CustomerID,
  o.CustomerID,
  o.OrderID
from `e1.customers` as c
left join `e1.orders` as o
on c.CustomerID = o.CustomerID
```

```sql
where o.OrderID is null




For every customer, show the number of orders they have placed,
including customers with zero orders.


select
  c.CustomerID,
  c.Name,
  count(o.OrderID) as no_of_orders
from `e1.customers` as c
left join `e1.orders` as o
on c.CustomerID = o.CustomerID
group by c.CustomerID, c.Name
order by no_of_orders asc


-- using right join


select
  c.CustomerID,
  c.Name,
  count(o.OrderID) as no_of_orders
from `e1.orders` as o
right join `e1.customers` as c
on c.CustomerID = o.CustomerID
```

```
group by c.CustomerID, c.Name
order by no_of_orders asc


For every customer, calculate:

Customer ID
Customer name
Number of orders placed
Total amount paid through successful payments only

Display only customers whose successful payment amount is
greater than ₹50,000.

Sort customers from highest to lowest successful payment amount.


select

from `e1.customers` as c
inner join `e1.orders` as o
on c.CustomerID = o.CustomerID
inner join `e1.payments` as p
on p.OrderID = o.OrderID


DPP - Advanced Questions

Find the top 10 products by total sales value.
```

Display:

ProductID
ProductName
CategoryName
SupplierName
Number of times the product appeared in order items
Total quantity sold
Total sales value
Average selling price

Only include products where the total sales value is greater than ₹1,00,000.

Consider only products having an MRP greater than ₹20,000.

For every category, calculate:

CategoryID
CategoryName
Number of qualifying products
Average MRP of qualifying products
Maximum MRP
Minimum MRP

Display only categories having more than 2 qualifying products.

Sort by average MRP from highest to lowest.

Find all customers who have never placed an order.

Display:

CustomerID
Customer name
City
State

Sort the result alphabetically by customer name.

For every supplier, calculate:

SupplierID
SupplierName
Number of products supplied
Average MRP of their products
Total sales generated by their products
Total quantity sold

```
Include suppliers even if their products have never been sold.

For suppliers with no sales, display the sales and quantity as 0.

Sort suppliers by total sales from highest to lowest.
```